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Review

The Role of Botulinum Toxin for Masseter Muscle Hypertrophy: A Comprehensive Review

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Abstract: Masticatory muscle hypertrophy (MMH) is a rare clinical phenomenon of uncertain etiology, characterized by a soft swelling near the angle of the jaw. This abnormal enlargement of the masseter muscle can alter the facial profile, leading to aesthetic concerns. Moreover, MMH may also have significant functional repercussions, including pain in the masseter region, often associated with temporomandibular disorders, fatigue, and discomfort during mastication. Non-conservative approaches offer an effective and minimally invasive solution by inducing localized muscle relaxation and reducing hypertrophy. Botulinum neurotoxin type A (BoNT/A) represents a therapeutic option for managing MMH, considering that injections can effectively reduce the masseter muscle volume, improving both facial aesthetics and related symptoms. Currently, the standard non-surgical management of MMH is BoNT/A injections, although consensus on the average dosage has not been definitely reached; on the other hand, there are data available in the literature about the injection technique of BoNT/A for lower face contouring. Therefore, the present comprehensive review aimed at exploring in detail the role of BoNT/A in the treatment of masseter muscle hypertrophy, describing its mechanism of action, the administration protocols, the clinical effects, and any side effects.

Keywords: masseter muscle hypertrophy; lower face contouring; botulinum toxin type A

1. Introduction

The masseter muscle is one of the primary muscles of mastication and plays a crucial role in mandibular movement and masticatory function. However, under certain circumstances, there can be an uncommon condition characterized by an abnormal increase in its size, defined as "masticatory muscle hypertrophy" (MMH) [1]. MMH has a high incidence in the second and third decades of life, with no sex predilection, can occur unilaterally or more commonly bilaterally (in about 60% of cases) [2,3], and although the condition is generally benign, can significantly impact both aesthetics and function [4]. Functional problems may manifest as protrusion of the jaw, pain, or headache [5]. The exact etiology remains unclear; however, it is most regarded as an adaptive response to chronic functional stimuli or repeated muscle stress [5]. Contributing factors such as masticatory habits—including excessive gum chewing, clenching, and bruxism—or genetic predispositions have been reported [6]. When no specific etiological factors can be identified, an idiopathic origin should be considered (IMMH) [7]. Notably, certain conditions such as psychological disorders or emotional disturbances for developing IMMH have been identified; these risk factors alter proprioception and affect the ability to maintain proper tone in the masseter muscle [2].

From an aesthetic perspective, MMH can significantly alter the facial profile by enlargement of the masseter and/or temporalis muscles, often leading to functional and aesthetic concerns. The increase in muscle volume can give the mandible a squared and angular appearance, affecting the perception of facial symmetry and harmony. This aesthetic effect often concerns patients, prompting them to seek treatments to improve their facial appearance [8]. MMH can have significant functional repercussions. Patients may report localized pain in the masseter region, often associated with temporomandibular disorders (TMDs). Patients may also experience fatigue or discomfort during mastication due to increased muscle tension and altered biomechanics [2].

This condition can impair masticatory efficiency and contribute to worsening TMDs, further exacerbating discomfort or chronic muscle tension [9–11]. In some cases, excessive muscle activity can impair masticatory function, causing difficulty in mandibular movements and contributing to bruxism [4]. Diagnosis of unilateral MMH could be made based on clinical examination and radiological findings [2]. The treatment of MMH varies depending on the severity of the condition and the patient's needs. Conservative treatments—such as the use of occlusal splints or night guards, oral devices to reduce muscle activity during sleep, physiotherapy as muscle massages and relaxation techniques to alleviate muscle tension, and behavior modification—aim to alleviate symptoms by reducing muscle hyperactivity and stress such as teeth grinding and excessive chewing of gum [6,12,13]. Non-conservative approaches, including botulinum toxin injections, offer an effective and minimally invasive solution by inducing localized muscle relaxation and reducing hypertrophy [14]. More in detail, botulinum neurotoxin type A (BoNT/A) represents a known therapeutic option for managing MMH considering that injections can effectively reduce the masseter muscle volume, improving both facial aesthetics and related symptoms [6,8,14,15]. The effect duration is temporary, requiring repeated treatments, but it offers a minimally invasive solution with a low risk of complications. Botulinum toxin type A (BoNT/A) injections are currently the standard non-surgical treatment for masseter muscle hypertrophy (MMH); however, a definitive consensus on the average dosage has

yet to be established. Nevertheless, the literature provides substantial data on BoNT/A injection techniques for lower face contouring [14,16,17].

Therefore, the present comprehensive review aimed at exploring in detail the role of BoNT/A in the treatment of masseter muscle hypertrophy, describing its mechanism of action, the administration protocols, the clinical effects, and any side effects.

2. Methods

Articles were selected through a comprehensive literature search in databases such as PubMed, Scopus, and Web of Science. The search strategy included keywords such as "botulinum toxin", "masseter muscle hypertrophy", "mechanism of action", "side effects", and "treatment outcomes." Inclusion criteria focused on studies published in peer-reviewed journals, primarily in English, and addressing the clinical applications, mechanism of action, or side effects of the botulinum toxin in masseter hypertrophy. Exclusion criteria included case reports and studies with insufficient data.

3. Masseter Muscle Hypertrophy

MMH is a rare clinical phenomenon of uncertain etiology [18], characterized by a soft swelling near the angle of the jaw, which may be associated with facial pain, significantly impacting facial aesthetics and functionality [19]. It can affect one side or both sides of the face and is characterized by an increase in the volume of muscle tissue, resulting in a change in the contour of the face [20]. In the literature, MMH is often described as a condition concomitant with the hypertrophy of other masticatory muscles, such as the temporalis muscles [21] or even medial pterygoid muscles [22]. Bilateral widening of the masseter muscles is more common but constitutes a minor aesthetic problem while maintaining facial symmetry [21]. Although the exact etiology of acquired hypertrophy of the masseter muscle is not well-known, there are several factors associated with its development, including the following: bruxism, TMD, pain, emotional stress, and oral parafunctions, including excessive unilateral chewing [6,22–24]. A recent study has demonstrated that intensive gum chewing can increase the stiffness of the masseter muscle and lead to bilateral hypertrophy of the masseter [25]. On the other hand, findings from several researchers suggest that the increase in muscle size is not caused by work-related hypertrophy but is the result of compensatory enlargement due to the lack of a specific type of muscle fiber [26]. Furthermore, along with the increase in muscle mass, changes may occur in the adjacent bone tissue, such as a thickening of the cortex of the angle and ramus of the mandible, the temporal fossa, and the zygomatic arch with a corresponding decrease in the area of the medulla or prominent exostoses at the angle of the mandibular bone, visible on computed tomography (CT) scans [27]. Bilateral enlargement of the masseter muscles is often accompanied by pain, which may be intermittent and may be confused with pain arising from the parotid gland [28,29]. Clinical examination usually reveals a mass of soft tissue near to the angle of the jaw, which becomes more prominent when clenching the teeth [30]. A limitation of mouth opening has been reported in some cases, particularly when the muscles are focally dystonic with tension in the region of the hypertrophic muscle [31]. It should be noted that the composition of muscle fibers in the enlarged masseter is very different from similar "work hypertrophy" muscles and healthy masseter muscles [32], suggesting that the term "hypertrophy" may be potentially misleading. Most patients report a slow but progressive nature of the condition [22,27]. The pathophysiology of muscle tone in patients with masseter muscle hypertrophy has not been studied. The increase in masticatory muscle tone is related to the increase in masticatory forces, generated by the masticatory muscles [14]. Both parameters increase negatively during parafunction, which means harmful movement habits (such as grinding, clenching, or chattering teeth, biting nails,

pressing the tongue against the teeth, and frequently chewing gum) that do not relate to physiological activities [33]. Stal et al. [34] pointed out that there is a different reactivity of masticatory muscles and skeletal muscles to stress because both these muscle groups have different embryonic origins; the masticatory muscles come from the mesenchyme of the first pharyngeal arch, while the skeletal muscles come from the mesoderm of the somite called the myotome, and Korfage et al. [35] highlighted the important advantage of diversity of fiber types, such as that observed in jaw muscles. Indeed, compared to the muscles of the limbs and trunk, the masticatory muscles possess a higher percentage of hybrid fibers, which express many subtypes of myosin or a greater number of type I fibers [36]. A large amount of hybrid fibers is associated with the plasticity of masticatory muscles and more efficient energy consumption during contraction [37]. Compared to the masticatory muscles, limb and trunk muscles have type II fibers with a larger diameter than type I. This suggests that the high presence of type I fibers with a small cross-sectional area (CSA) in masticatory muscles could facilitate the greater exchange of nutrients and O_2 with the extracellular environment, increasing the fibers' resistance to fatigue [38]. Excessive and prolonged muscle tension becomes the cause of non-physiological and excessive overload of the TMJ, which in turn leads to damage to the soft tissue elements inside the joints [39].

4. Botulinum Toxin: Mechanisms of Action and Indications

Botulinum toxin is a neurotoxin produced by the anaerobic bacterium *Clostridium botulinum*, consisting of a neurotoxic core and non-toxic proteins. There are seven serotypes of BT (A, B, C, D, E, F, and G), each of which has additional subtypes (e.g., subtype A contains several distinct subtypes). The growing use of botulinum neurotoxins (BoNTs) in medical and aesthetic applications has spurred the development of various BoNT products [40]. All serotypes have a similar chemical structure and are neurotoxins, except for subtype C2, which belongs to the family of binary AB-type toxins, structurally composed of separate enzymatic (A, C2I) and binding (B, C2II) components [41].

Among its seven serotypes, the BoNT/A is the most extensively studied and clinically applied, particularly in aesthetic medicine, oncology, neurology, rehabilitation, and dentistry [42–45]. Differences among the botulinum toxin serotypes lie in their specific protein targets and clinical applications. For instance, BoNT/A and BoNT/B target distinct proteins in the SNARE complex; while BoNT/A cleaves the SNAP-25 protein, BoNT/B specifically targets synaptobrevin (VAMP), another critical component of the vesicle fusion machinery. This distinction contributes to their unique pharmacological profiles. BoNT/A is the primary choice for aesthetic and neurological applications due to its well-characterized safety and efficacy, whereas BoNT/B is sometimes preferred in cases of resistance to BoNT/A, such as in cervical dystonia, where its alternative protein target may overcome immunogenicity issues [45–47].

Upon injection, BoNT/A is endocytosed into the cell and cleaves the C-terminal portion of SNAP-25. This cleavage disrupts the SNARE complex, blocking the exocytosis of neurotransmitters and the acetylcholine release [48]. Additionally, BoNT/A suppresses the peripheral release of neuropeptides and inflammatory mediators, including CGRP (calcitonin gene-related peptide), which plays a key role in pain signaling and inflammation [49–51]. The reduction in CGRP in the peripheral and central nervous system is thought to contribute to its therapeutic effects in migraine management [41,52,53]. CGRP release occurs in the meninges and the trigeminal caudal nucleus and is amplified by P2Y purinergic receptors on satellite glial cells [54], which are influenced by factors such as 17-beta-estradiol. Conversely, P2X3 receptor-mediated peripheral pain signaling in neuronal afferents can attenuate CGRP activity [55]. Furthermore, receptors such as the transient receptor potential vanilloid 1 (TRPV1) play a critical role in pain perception

and sensitization. These receptors may represent additional therapeutic targets for pain management [56]. The toxin consists of a 100 kDa heavy chain (HC) and a 50 kDa light chain (LC) linked by a disulfide bond. The heavy chain facilitates receptor binding and endocytosis into the neuron, while the light chain cleaves SNAP-25—a component of the SNARE complex—to inhibit acetylcholine release at the neuromuscular junction. This process leads to a reversible flaccid paralysis, a key therapeutic mechanism in conditions like spasticity, dystonia, and migraine.

BoNT/A has demonstrated significant effects in treating MMH, bruxism, TMDs, and oromandibular dystonia, especially in terms of function recovery [57]. Indeed, intramuscular injections into the masseter and anterior temporalis muscles reduce muscle tone and bulk, alleviating pain and improving jaw functionality. The therapeutic effect of intramuscular injections depends on the section of the muscle, the dose administered, and the injection method [58,59]. Electromyographic studies confirm the decreased neuromuscular activity post-injection, with effects lasting 3–4 months [49,59]. Additionally, the reduced formation of axonal bundles during recovery promotes sustained muscle relaxation, enhancing therapeutic outcomes [60]. In neurology, BoNT/A has become a cornerstone therapy for chronic migraine. By targeting CGRP release in the trigeminal system and modulating pain transmission pathways, the toxin offers a dual benefit—reducing nociceptive signaling and peripheral sensitization. Emerging studies also suggest that receptors like TRPV1 and purinergic P2X3 may represent complementary targets in pain management, further broadening the therapeutic potential of BoNT-A [52,56].

5. How to Treat a "Square Face": Role of Botulinum Toxin

BoNT/A is the most popular subtype for injection for aesthetic purposes. To date, botulinum toxin has been widely and mainly used for the treatment of conditions affecting the upper and middle face; however, due to recent efforts and patients' needs, the indications for cosmetic reasons of botulinum toxin injection have been expanded to the lower face and neck areas [59]. The use of botulinum toxin for the aesthetic contouring of the lower face is extending, and it is typically achieved by injecting the masseter muscle.

The most common cosmetic complaint for which patients seek treatment is a "square face", and the solution is the injection of botulinum toxin into the abovementioned muscle to decrease muscle hypertrophy. A triangular-shaped lower part of the face with a perfect jawline is the most desired shape by women compared to a wide and square one, which for many is considered more attractive in males. The aesthetics of the lower third of the face is determined by three main factors: size of the bony jaw, mass of the masseter muscle, and volume of the subcutaneous adipose tissue [60]. The size and shape of the jaw are predetermined at birth and are fixed at adulthood; these factors also vary between different ethnicities and races. The volume of subcutaneous fat varies from person to person based on their individual body mass index and can change over time, according to general health and body fat percentage [60]. Masseter hypertrophy can result in a wide jaw, which negatively affects the aesthetic appearance of patients [61]. The procedure is performed by injection with a 30 G needle inserted deep into the muscle; the injection safety zone can be easily determined. The inferior border of the mandible is marked, which is the lower margin of the injection zone; the upper margin is a line drawn from the corner of the mouth to the earlobe. The patients are asked to grit the teeth so that the anterior and posterior borders of the masseter muscle can be palpated and marked. Thus, a rectangle shape is drawn on the lower face, and the injection safety zone is depicted within this rectangle, 1 cm from the margins [61]. Operating within this area, recommendations most commonly suggest three injection points, using a total amount of 24 to 60 U of onabotulinumtoxinA or incobotulinumtoxinA and 60 to 300 U of abobotulinumtoxinA [60,61]. Chang et al. [62]

have reported a favorable long-term efficacy for this method, with a 12% reduction in masseteric muscle volume after three consecutive injections, 1 year after completion of the treatment. The minimum dosage used in a previous study for toxin injection into the masseters was 20 U of onabotulinumtoxinA on each side with two injection sessions at a 4-month interval [63]. The standard non-surgical treatment for MMH currently involves BoNT/A injections; however, there is no definitive consensus on the average dosage. Nevertheless, the existing literature provides valuable insights into the injection techniques of BoNT/A for achieving lower face contouring [64–66].

While some authors argue that a multi-point injection allows for better distribution of the toxin into the muscle, single-point injections are quicker to perform and less painful, and may be less prone to producing complications [67]. In contrast, in 2018, Nikolis et al. [68] evaluated the effectiveness of two injection techniques in the treatment of 30 patients with MMH using the toxin incobotulinumtoxinA. Fifteen patients received 40U of incobotulinumtoxinA at a single injection point and fifteen patients received 40U with a 5-injection technique (8U for each injection). At the 6-month follow-up, they demonstrated no significant difference between the two injection techniques. It has been proven that botulinum toxin is a safe treatment, even after repeated sessions for several years. However, in most cases, there is a need for using a combination of botulinum toxin injection with other procedures to obtain optimal results [66,67]. An example of combination therapy would be fat grafting/autologous fat grafting with botulinum toxin injection into the masseters as a safe and satisfactory method for short and wide faces in order to normalize the ratios of different areas of the lower face and obtain a desirable oval face [69].

6. Role of Botulinum Toxin for Masseter Muscle Hypertrophy

Recently, the scientific literature has revealed a higher interest regarding the effectiveness and safety of BoNT/A, highlighting its potential as a non-invasive therapeutic option in masseter muscle hypertrophy [70]. A recent review performed by Kundu et al. [60] analyzed the effect of BoNT/A in masseter muscle, stating that botulinum toxin was consistently effective in relieving pain at 3 and 6 months after treatment with an effect that is probably dose-dependent. BoNT/A was consistently effective in relieving pain 3 and 6 months after treatment [59,60]. It is well-demonstrated that for each unit of increase in the dose of BoNT/A, the severity of masseteric hypertrophy decreased in terms of pain [62,69,70]. A recent study has demonstrated that a single injection in masseter muscle seems to have a clinically subjective and objective effect on masseteric hypertrophy [69,70]. Furthermore, patients who received three injections experienced a higher efficacy and longer duration of treatment maintenance compared with those who received two injections [71]. Several techniques of injecting BoNT/A into masseter muscle have been employed in the past. Some authors are used to the multi-point injection technique since it allows for a better distribution of the toxin in the muscle, while some use the lowest injection sites possible or a single injection site [69]. Moreover, several studies demonstrated the efficacy of a biphasic injection method that is associated with minimum self-resolving side effects [72–75]. The authors proved the role of extensive knowledge of muscular anatomy and an appropriate injection technique as key factors in desirable clinical outcomes and decreasing side effects [75]. Nevertheless, comparing the two methods of multi-injection and single-injection treatments, no superiority of either method was found. Moreover, it has been suggested that the analgesic effect of BoNT/A could involve the reduction in neurotransmitter release at both peripheral and central levels [73]. Consequently, BoNT/A may have a direct analgesic effect by blocking the transmission of pain signals from the temporomandibular area to the nervous system [74]. This might be due to BoNT/A's ability to inhibit the release of certain neurotransmitters involved in pain signaling [60].

From a rehabilitative perspective, a recent systematic review of 2024 affirms that BoNT/A can improve the masseter hypertrophy and hypertonicity, significantly alleviating joint stress and improving jaw function [68]. Furthermore, several RCTs support its use for treating TMDs that can arise as a consequence of bruxism or independently [75,76]. The long duration of action of BoNT/A for pain relief, which can persist up to 6 months after treatment, could be due to the ability of its protease to avoid cellular degradation mechanisms and survive in the cell cytoplasm for a prolonged period [60]. In fact, it has been hypothesized that the pain-relieving effect of BoNT/A persists even after the return of muscle strength [77]. A potential explanation could also be that bruxists became more aware of parafunctional (teeth clenching) habits and took corrective measures, which may have restored the normal function of other masticatory muscles [78]. In this context, muscle thickness of the masseter muscle can play a pivotal role in orofacial pain. The use of BoNT/A can decrease the muscle's thickness by an average of 31% after a 3-month follow-up, as measured by ultrasound, and volume loss continues even after restoration of the muscle function [67,79–81]. Recent studies showed a decrease in muscle thickness and an improvement in masseter contraction ability and masticatory performance. The authors conclude that a single injection of BoNT/A does not seem to affect muscle thickness—assessed by ultrasound—permanently since the thickness had recovered after 3 months [69]. However, the muscle's thickness following a booster injection 3 months after the first did not return to normal after 6 months [69]. Previous animal studies and some human studies indicate that this could be a result of incomplete re-innervation of the injected area, fatty infiltration, fibrosis, and even atrophy due to necrosis of the muscle fibers in mice and humans [82]. It is interesting to note that the evaluation of muscle thickness using a caliper has shown that muscle thickness is directly proportional to the symptoms reported by the patient: after the injection, the thickness of the masseter decreases and, similarly, the symptoms disappear or are reduced [67]. Moreover, facial aesthetics also improve thanks to the reduction in thickness and muscle relaxation [83]. It is important to note that muscle thickness is often correlated with muscle strength. Thus, clinicians normally make dose adjustments based on the hypertrophy of the masseter muscle while avoiding the prevalence of side effects [84]. Moreover, it has been stated that the human system produces antibodies when exposed to excessive BoNT/A, which may impair its effectiveness in future treatment but, to date, BoNT/A has shown a very low protein content, avoiding this limiting effect [85–87]. To achieve personalized dose injections based on the ultrasound evaluation of masseter muscle and deep tendon thickness, Shi et al. [88] assessed the morphological changes in the masseter muscle and deep tendon of 206 subjects under relaxed and clenched conditions using ultrasound. The study found that the deep belly was thicker in both relaxed and clenched states in longitudinal ultrasound images; however, when Li et al. [18] used the same measurement method on 42 participants, they reached an opposite conclusion. Regarding muscle activity, compared to placebo injections, BoNT/A was associated with a significant decrease in occlusal force (kg) 3 months after injection, with a decrease in effects after 3 months and no significant differences at 6 months [89–92]. Comparing BoNT/A treatment with the use of occlusal bites and placebo injections, the analysis of maximum chewing force after BoNT/A injections demonstrated a significant reduction at 1 month or less. BoNT/A continued to outperform occlusal splints and saline placebo at 3 months [70]. Between 3 and 6 months, a significantly higher maximum chewing force was observed in the BoNT/A group compared to the oral splinting group [70]. No significant differences in maximal chewing force were observed between the BoNT/A and saline placebo groups [70]. BoNT/A works by controlling the intense contractions of the masticatory muscles [93]. Injection of the masseter alone is, therefore, sufficient to treat bruxism [94,95], with the other masticatory muscles (temporalis, pterygoids, digastrics, and genius-hyoids) capable

of compensating and, therefore, supporting effective chewing [96]. The favorable mechanism of neuromuscular conduction block significantly reduces the tone of the masseter muscle, as confirmed by electromyographic (EMG) evaluation [97,98]. Furthermore, the recovery of conductivity occurs based on the formation of new axonal bundles, which are in much smaller quantities than the original connections, a very favorable phenomenon for achieving muscle relaxation [99]. In one study, post-injection polysomnographic evaluation confirmed that botulinum reduces the intensity of contraction of the masseter muscle for at least 12 weeks but does not modify the frequency of occurrence of rhythmic masticatory muscle activity (RMMA) episodes. On the other hand, the author offers a different perspective of bruxism: RMMA episodes are physiological so it is not necessary to intervene to try to reduce them and improve symptoms [89,98]. The efficacy of botulinum toxin treatment on the range of joint movements was observed: compared to a placebo, BoNT/A was associated with a significant increase in maximum pain-free opening (mm) 1 month after injection, and in unassisted opening (mm) at 1 month and 6 months post-injection; no difference was observed in maximum unassisted opening (mm) at 1 week or 2 months post-injection [67]. Compared to the placebo, BoNT/A was associated with a significant increase in the right lateral excursion (mm) at 1 week, 1 month, and 6 months; and in the left lateral excursion (mm) at 1 week, 1 month, and 6 months [67]. However, according to Asutay et al. [99], changes in voluntary mouth opening did not show statistically significant differences, unlike what was studied by Sidebottom et al. [100] and Guarda-Nardini et al. [101]. Other studies agree on an improvement in temporomandibular muscle and joint function in patients with muscular TMD [102–108]. In fact, compared to placebo injections, BoNT/A is associated with a significant post-injection reduction in rest pain intensity at 1 month, 2 months, 3 months, and 6 months from the first injection [102–104]. Additionally, BoNT/A was found to be associated with a significant decrease in masseter muscle activity at 1 month and occlusal force at 3 months [90]. Furthermore, BoNT/A resulted in a significantly greater increase in maximum pain-free mouth opening, unassisted mouth opening, and mandibular excursive movement at 1 month post-treatment [102]. Furthermore, several systematic reviews assessed the use of BoNT/A therapy for myofascial pain, reporting a reduction in pain intensity in the BoNT/A groups and concluding that this treatment was slightly more effective than placebo for pain reduction but appears as effective as other conservative and rehabilitative approaches [102–105]. Thus, some authors conclude that no strong recommendations can be drawn [106]. The therapeutic effect of intramuscular injections depends on the section of the muscle, the dose administered, and the injection method but the literature agrees on the duration of effects ranging from 3 to 4 months [59,72]. In summary, injection of a low dose of BoNT/A seems to have a clinically subjective and objective effect on masseteric hypertrophy. However, as a clinician, one must consider the possible harms that can occur with repeated injections in shorter intervals, which include impaired masticatory performance and muscle activity but also a permanent decrease in muscle thickness with consequent functional loss [107]. In comparison, the effects of BoNT/A generally last 3–6 months for aesthetic facial lines [108]. The longer duration of effectiveness after treatment of the masseters compared to treatment of the facial lines may be related to dosage [109].

7. Protocols of Botulinum Toxin Injections for Masseter Muscle Hypertrophy

Currently, the standard non-surgical management of MMH is BoNT/A injections, although a consensus on the average dosage has not been definitely reached; on the other hand, there are data available in the literature about the injection technique of BoNT-A for lower face contouring [14,16,17].

While some authors suppose that an injection in multiple injection sites allows for a better distribution of the toxin in the muscle and ensures a uniform reduction in the size of the masseter [102], single-point injections are quicker to perform and less painful, and may be less prone to producing complications. The data, therefore, suggest using as few injection sites as possible until more robust evidence is presented [110]. The most important concept when injecting BoNT/A into the masseter is to stay well within the confines of the muscle to avoid complications [62]. The ideal injection site is the most prominent point of the swollen masseter muscle, where the appropriate nerve innervates the affected muscle [66,111]. The number of BoNT/A injection sites and dosages may depend on the thickness of the masseter muscle and the degree of hypertrophy [66]. The superior portion of the masseter muscle is considered unsuitable for BoNT/A injection due to the location of the parotid duct and the absence of branches of the perforating nerve supplying the superficial layer of the masseter muscle [111]. To ensure proper targeting, clinicians can ask patients to clench their teeth, which facilitates the identification of the hypertrophic muscle [108]. A safe zone of the muscle lies below the line connecting the earlobe and the corner of the mouth, and between the posterior margin of the muscle and 1 cm after the corner of the mouth. Within this zone, most recommendations suggest using three injection points with a total dose of 24–60 units of onabotulinumtoxinA or incobotulinumtoxinA, or 60–300 units of abobotulinumtoxinA, delivered via a 30-gauge needle at appropriate depths into the muscle [110]. Li et al. [102] analyzed injection sites and the results indicated that at 6 months post-injection, BoNT/A injections to the masseter, temporalis, and pterygoid muscles were associated with greater pain reduction, while injections to the masseter muscles alone or masseter and temporalis muscles produced a smaller effect size. This suggests that broader targeting (masseter, temporalis, and pterygoid muscles vs. masseter and temporalis) may provide superior outcomes for pain management [102]. Despite the variability in dosing protocols, most studies agree that doses lower than 20 units are inadequate for significant results, while higher doses (20–40 units) yield better effects, particularly in patients with severe symptoms. The U.S. Food and Drug Administration (FDA) has approved a maximum dose of 400 units of onabotulinumtoxinA or incobotulinumtoxinA within a three-month interval. However, doses up to 600 units may not significantly increase the risk of adverse events. The neurotoxin load (in ng per 100U) varies among BoNT-A formulations: 0.73 ng/100U for onabotulinumtoxinA; 0.65 ng/100U for abobotulinumtoxinA; and 0.44 ng/100U for incobotulinumtoxinA. Consequently, the specific potency of the 150 kDa BoNT-A neurotoxin is estimated at 137 units/ng for onabotulinumtoxinA; 154 units/ng for abobotulinumtoxinA; and 227 units/ng for incobotulinumtoxinA [112]. A summary of the literature about safety and maximum doses is reported in Table 1.

Table 1. Injection techniques and doses for masseter muscle hypertrophy.

Injection Technique	Number of Injection Sites	Total Dose	Toxin Type	Notes (with References)
Multi-point injection	3–5	24–60 U (onabotulinum/incobotulinum toxin A)	Onabotulinum/ Incobotulinum	Ensures uniform toxin distribution within the muscle and reduces asymmetry risks [91,112]
Single-point injection	1	20–40 U (onabotulinum toxin A)	Onabotulinum	Faster and less painful; may result in uneven muscle reduction [108,113,114]
Ultrasound-guided injection	Variable	20–60 U	Onabotulinum/ Incobotulinum	Increases precision, reduces diffusion risks, and ensures accurate delivery [62,91,108]
Multi-muscle protocol (masseter, temporalis, and in some cases, the pterygoid muscles)	Variable	20–60 U (masseter, temporalis, pterygoid)	Onabotulinum	Associated with significant pain reduction in TMD patients [71]

Table 1. *Cont.*

Injection Technique	Number of Injection Sites	Total Dose	Toxin Type	Notes (with References)
Abobotulinum toxin A	3	60–300 U	Abobotulinum	BoNT-A neurotoxin is estimated at 137 units/ng for onabotulinumtoxinA, 154 units/ng for abobotulinumtoxinA, and 227 units/ng for incobotulinumtoxinA [112–114]
High-dose injection	Variable	Up to 400 U every 3 months	All types	Effective for severe TMD; no significant increase in adverse events with higher doses [14,115]
Repeat treatment	Variable	15–30 U per side	Onabotulinum	Muscle thickness and dosage decrease with repeated injections over time [116]
Combination therapy (masseter + others)	3+	60–100 U (masseter, temporalis, pterygoid)	Onabotulinum	Combined improvement in TMD symptoms and facial contouring [71,92]

The first reported case of MMH treatment, described by Moore and Woode in 1994 [14], used 300 units of abobotulinumtoxinA in a 30-year-old man with TMJ dysfunction, myofascial pain syndrome, and masseter hypertrophy. The patient exhibited a slight muscle reduction for two weeks post-injection, with effects lasting six months without side effects. One of the largest studies, conducted by Kim et al. [70], involved 1021 patients treated with 100–140 units of abobotulinumtoxin A (ABO) for lower face contouring (383 patients could be followed for >3 months). The thickness was reduced by 31% and the maximum reduction in muscle strength was observed after 10–12 weeks and 50% required a second injection after 4–7 months. Then, the large-scale trial by Kim et al. [70] reported consistent success using lower doses, indicating that modern approaches are increasingly effective with reduced toxin volumes. A reduction in symptoms can be achieved with doses less than 25U applied exclusively to the masseter muscles [97]. However, recommendations most commonly suggest using a total amount of 24 to 60 U of onabotulinumtoxinA or incobotulinumtoxinA and 60 to 300 U of abobotulinumtoxinA at three sites, with a 30 G needle inserted deep into the muscle [117]. A higher bilateral dose of 60–100 U could instead be the optimal choice for the treatment of muscle TMD pain. Notably, a higher dose of BoNT/A resulted in greater pain reduction at 6 months compared to a lower dose [102].

8. Side Effects, Complications, and Contraindications of Botulinum Toxin for Masseter Muscle Hypertrophy

8.1. Side Effects of Botulinum Toxin for Masseter Muscle Hypertrophy

Botulinum toxin injections into the masseter muscle are widely regarded as safe and effective for both therapeutic and aesthetic applications. BoNT/A has proven to be a safe treatment, even after repeated sessions for several years [118]. All studies focus on identifying a safe injection area rather than on the specifics of the technique [111]. There were favorable results by selecting the locations and depth of BoNT/A applications by clinical estimation and palpation, with few or no reports of adverse effects [97]. The safe injection area is usually delimited by the lower mandibular border, the margins of the anterior and posterior masseter, and the zygomatic arch as the upper limit. Various side effects (intended as predictable, generally mild, and often dose-dependent outcomes of treatment) associated with the procedure have been reported, most likely secondary to excessive infiltration or inaccurate placement of the BoNT/A, albeit up to 50% of patients may experience adverse events (as harmful and unintended outcomes, which may range from mild to severe and may not always be directly linked to the treatment), which are

mostly mild, transient, and self-resolving within 1–2 months [119]. Table 2 depicts the main side effects of BoNT/A for MMH management [51,71,77,82,107,118,120–123].

Table 2. Side effects of botulinum toxin treatment for masseter muscle hypertrophy.

Incidence	Side Effects	Management
Common (~30–50%)	Mild pain at injection site, temporary muscle weakness	Analgesics, reassurance; dose adjustment
Occasional (~2–15%)	Facial asymmetry, bruising, headache, altered chewing mechanics, sunken cheeks	Precise injection techniques; local compression; analgesics; tailor dosage to individual anatomy
Rare (~0.2–2.3%)	Incipient sagging of cheeks/skin, sagging of jaw soft tissues	Use lower doses; distribute sessions; consider depressor muscle injections
Rare (~<1–2%)	Aesthetic changes in smile, irregular swelling of muscles, speech disorders, herniation of parotid gland, bone remodeling changes	Inject within masseter boundaries; employ ultrasound guidance; monitor and adjust treatment
Very Rare (<1%)	Immune-mediated reactions	Cease treatment; consult immunologist

The most common complications of injections (as unintended and often preventable deviations from the expected course of treatment) include mild pain at injection sites or temporary muscle weakness (e.g., atonia of the injected muscle), which can lead to unwanted weakness or functional limitations such as reduced chewing strength and difficulty opening the mouth [124]. Aesthetic concerns include asymmetry of the smile—particularly in facial applications—problems opening the mouth, bruising (accounting for 2.5%), incipient sagging of the cheeks and skin, local pain at the injection site, mild headache, awkward facial expressions, speech disorders, irregular swelling of the muscles, sunken cheeks, and herniation of the parotid gland through the overlying attenuated muscle. After treatment, due to the rapid reduction in size of the masseter muscle, sagging of the skin and soft tissue in the jaw area may occur in 0.2% to 2.3% of patients, resulting in a blurred and more aged appearance of the jaw [75]. These are particularly common in older patients or those with thinner skin as rapid masseter muscle atrophy can lead to skin laxity of the soft tissues of the mandibular border and a more aged appearance [14]. This is mainly caused by gravity and the tensile force of the platysma muscle [72]. Treatment strategies to mitigate these side effects include lower doses and spaced sessions as well as complementary injections into depressor muscles to counteract sagging [65]. These effects are often dose-dependent, localized, and uncommon [107] and can be minimized by employing precise injection techniques and tailoring the dosage to individual patient anatomy and needs [51,125]. Furthermore, since the effect of BoNT/A is reversible, muscle weakness gradually recovers over the course [84]. Functional side effects include transient facial asymmetry and altered chewing mechanics, which can result from unintended weakening of adjacent muscles such as the risorius or zygomaticus. Two patients in the study by Kim et al. [70] reported aesthetic changes in their smile, which may be due to the diffusion of botulinum toxin toward the zygomaticus and risoris muscles during application. To avoid this, the BoNT/A must be injected well within the confines of the masseter muscle and, superiorly, a line joining the tragus to the oral commissure must be considered as the superior margin.

8.2. Complications of Botulinum Toxin for Masseter Muscle Hypertrophy

BoNT-A is widely regarded as a safe and effective therapeutic agent, yet rare complications, though infrequent, demand attention and careful management. Occasionally,

paradoxical muscle swelling may occur, presenting as localized bulging despite the intended reduction in muscle activity. This phenomenon is often linked to uneven toxin diffusion or anatomical barriers, such as the deep inferior tendon within the masseter muscle. Ultrasound guidance and carefully distributed injections are key to preventing such outcomes [126,127]. Other rare effects include dysphagia and speech difficulties resulting from unintended spread to the adjacent muscles, as well as transient facial nerve palsy causing asymmetry or weakness. These complications can be avoided with refined injection techniques and precise localization of the toxin [62]. Systemic toxicity, while extremely rare, underscores the importance of adhering to safe dosing practices and immediate intervention in case of overdose [119]. Long-term use of BoNT/A in masticatory muscles has raised concerns about mandibular bone thinning, particularly in animal studies. Several studies have examined the impact of BoNT/A injections on mandibular bone structure [123–127]. Moussa et al. [123] have recently reported, in a systematic review and meta-analysis, the adverse effects of BoNT/A injections on mandibular bone tissue. The authors reviewed both preclinical and clinical studies to assess the impact of BoNT/A dosage on bone density and mandibular structure. We recognize that the dose–response relationship is a critical factor in interpreting the findings of both preclinical and clinical studies. The author found that in animal studies, higher doses of BoNT/A were employed relative to muscle mass compared to the doses used in human studies. Conversely, human studies typically used doses within the therapeutic range (e.g., 20–50 units per side for masseter hypertrophy), which are less likely to result in significant bone changes. Therefore, the higher doses in animal models have demonstrated more pronounced effects on mandibular bone density and microarchitecture, while lower, therapeutic doses in humans appear to have minimal or no clinically significant impact. However, the authors concluded that while the preclinical evidence points to negative effects of BoNT/A on mandibular bone health, the clinical evidence remains insufficient to draw definitive conclusions [123]. They recommend caution with prolonged or high-dose BoNT/A use in masticatory muscles and emphasize the need for further clinical research to assess the long-term impact of these treatments on mandibular bone structure. Hong et al. [124] observed a reduction in cortical bone quality following BoNT/A injections, highlighting the potential for bone structure deterioration due to decreased mechanical loading. Research by Tsai et al. on rats [125] indicated that BoNT/A injections into masticatory muscles led to reduced cortical bone thickness in specific mandibular regions, such as the angular process, due to muscle atrophy and subsequently decreased mechanical forces. Kahn et al. [126] reported that BoNT/A-induced muscle atrophy resulted in decreased bone quality in the mandibular condyle and alveolar bones, emphasizing the relationship between muscle function and bone integrity. These studies collectively suggest that BoNT/A injections can lead to a reduction in mandibular bone quality, particularly in cortical thickness, due to the interruption of normal bone remodeling processes. The regions of interest commonly affected include the mandibular condyle, coronoid process, and angle/ramus, with significant changes observed within 3 to 12 months post-injection [126]. Therefore, mandibular bone loss could be considered an adverse effect of BoNT/A's application on masticatory muscles; even though more studies are needed, the results extrapolated to humans [102] highlight the need for caution in prolonged or high-dose treatments [126]. Allergic reactions such as allergic dermatitis or antibody formation, although rare, have been described in a small percentage of patients, while systemic reactions of flu-like symptoms or generalized malaise are exceedingly rare [127]. These reactions are usually mild and self-limiting [127]. Genetic predispositions and immune variability among patients may further influence susceptibility to these events [120]. Management strategies focus on using the lowest effective dose, extending intervals between treatments, and considering alternative formulations

when antibody resistance is suspected [120]. Huang et al. [66] demonstrated that three-dimensional photography combined with ultrasound could track volume reduction more accurately, enabling better patient satisfaction and tailored treatments. The injection of botulinum toxin into the orofacial region can also be guided with the help of electromyography [110,126]. To reduce the incidence of rare complications associated with BoNT/A treatments, practitioners can employ several preventative strategies focusing on precision, technique, and patient-specific factors, including the use of ultrasound-guided injections.

To mitigate immune-mediated responses, it is crucial to minimize cumulative doses and extend the intervals between treatments. For patients who develop resistance due to neutralizing antibody formation, switching to alternative toxin formulations may restore efficacy [121]. Finally, patient education cannot be overlooked. Informing patients about potential transient effects, such as mild chewing weakness or temporary facial asymmetry, prepares them for recovery. Advising rest post-treatment can further help reduce minor side effects like dizziness or headaches. With a combination of advanced guidance tools, careful injection planning, personalized treatment approaches, and patient education, clinicians can significantly reduce the risk of rare complications. These strategies ensure safer and more effective outcomes in botulinum toxin therapies [69,93,118,124,127]. Table 3 depicts the risks and mitigation methods in detail.

Table 3. Risks and mitigation methods for botulinum toxin treatment.

Risk	Mitigation Method
Facial asymmetry	Ensure proper pre-treatment assessment and dose adjustments; inject within safety zones.
Chewing weakness	Use the lowest effective dose; avoid excessive infiltration; use ultrasound or EMG guidance.
Bruising at injection site	Apply local compression immediately post-injection; use smaller-gauge needles.
Sagging skin and soft tissues	Lower injection doses; distribute treatment over several sessions; complementary injections in depressor muscles.
Paradoxical swelling	Use ultrasound guidance to confirm masseter anatomy; superficial booster doses may help if swelling persists.
Diffusion to unintended muscles (e.g., risorius)	Ask patients to clench their teeth during injection to identify muscle boundaries; inject well within masseter confines.
Bone changes (e.g., osteopenia)	Avoid prolonged or high-dose treatments; monitor long-term effects; further research needed for validation.
Unnatural facial expressions	Inject at appropriate depth and within masseter boundaries; adjust doses based on patient anatomy.
Pain at injection site	Use precise injection techniques and ensure proper patient positioning.

Legend: This table outlines the key risks associated with botulinum toxin treatment and the corresponding mitigation methods. The risks range from mild, transient effects like bruising and pain at the injection site to more significant complications such as facial asymmetry, sagging skin, or unintended diffusion of the toxin. Mitigation strategies include precise injection techniques, ultrasound guidance, dose adjustments, and appropriate pre-treatment assessments to ensure patient safety and optimal therapeutic outcomes.

9. Limitations of This Study

The present study has the main limitation of being a narrative review. Thus, this study's design did not allow us to provide evidence according to systematic research of the scientific literature, nor did it provide quantitative conclusions. Additionally, variability in the designs, sample sizes, and methodologies across the included studies is a limiting factor. Furthermore, the lack of randomized controlled trials comparing botulinum toxin protocols for masseter hypertrophy represents another limitation.

10. Conclusions

In summary, BoNT/A injections in the masseter muscle are a valuable tool for both therapeutic and aesthetic purposes. The treatment of MMH with botulinum toxin requires a careful balance of dosage, technique, and frequency to achieve both functional and aesthetic improvements. While significant advancements have been made in refining protocols, the variability in each patient's anatomy and needs underscores the importance of individualized treatment planning. BoNT-A is effective and safe in the treatment of MMH and TMDs, offering significant benefits in terms of pain reduction, improvement in muscle and joint function, and aesthetic advantages, with effects lasting several months. While most side effects are mild and self-limiting, rare complications underscore the importance of precision, advanced techniques, and individualized treatment planning. By utilizing innovations like ultrasound-guided injections, clinicians can further enhance the safety and efficacy of these treatments. Continued research into the long-term effects of BoNT/A, particularly on mandibular bone health, is essential to ensure optimal patient outcomes. Future studies are needed to establish more definitive guidelines for optimal dosing, injection sites, and treatment intervals and to better understand the mechanisms underlying botulinum toxin's effects on muscle function.

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References

1. Smyth, A.G. Botulinum toxin treatment of bilateral masseteric hypertrophy. *Br. J. Oral Maxillofac. Surg.* **1994**, *32*, 29–33. [CrossRef]
2. Gurney, C.E. Chronic bilateral benign hypertrophy of the masseter muscles. *Am. J. Surg.* **1947**, *73*, 137–139. [CrossRef]
3. Baek, S.M.; Kim, S.S.; Bindiger, A. The prominent mandibular angle: Preoperative management, operative technique, and results in 42 patients. *Plast. Reconstr. Surg.* **1989**, *83*, 272–280. [CrossRef] [PubMed]
4. Xie, Q.; Yang, C. Masticatory muscle hypertrophy: Etiology and management strategies. *Oral Dis.* **2018**, *24*, 13–20.
5. Rispoli, D.Z.; Camargo, P.M.; Pires, J.L., Jr.; Fonseca, V.R.; Mandelli, K.K.; Pereira, M.A.C. Benign masseter muscle hypertrophy. *Braz. J. Otorhinolaryngol.* **2008**, *74*, 790–793. [CrossRef] [PubMed]
6. Fedorowicz, Z.; van Zuuren, E.J.; Schoones, J. Botulinum toxin for masseter hypertrophy. *Cochrane Database Syst. Rev.* **2013**, *2013*, CD007510. [CrossRef]
7. Singh, S.; Shivamurthy, D.M.; Agrawal, G.; Varghese, D. Surgical management of masseteric hypertrophy and mandibular retrognathism. *Natl. J. Maxillofac. Surg.* **2011**, *2*, 96–99. [PubMed]
8. Srivastava, S.; Kharbanda, S.; Pal, U.S.; Shah, V. Applications of botulinum toxin in dentistry: A comprehensive review. *Natl. J. Maxillofac. Surg.* **2015**, *6*, 152–159.
9. Ferrillo, M.; Gallo, V.; Lippi, L.; Bruni, A.; Montrella, R.; Curci, C.; Calafiore, D.; Invernizzi, M.; Migliario, M.; de Sire, A. The 50 most-cited articles on temporomandibular disorders: A bibliometric analysis. *J. Back Musculoskelet. Rehabil.* **2023**, *36*, 279–297. [CrossRef]
10. Ferrillo, M.; Migliario, M.; Marotta, N.; Fortunato, F.; Bindi, M.; Pezzotti, F.; Ammendolia, A.; Giudice, A.; Bonda, P.L.F.; de Sire, A. Temporomandibular disorders and neck pain in primary headache patients: A retrospective machine learning study. *Acta Odontol. Scand.* **2023**, *81*, 151–157. [CrossRef] [PubMed]

11. Ferrillo, M.; Marotta, N.; Giudice, A.; Calafiore, D.; Curci, C.; Fortunato, L.; Ammendolia, A.; de Sire, A. Effects of Occlusal Splints on Spinal Posture in Patients with Temporomandibular Disorders: A Systematic Review. *Healthcare* **2022**, *10*, 739. [CrossRef]
12. Guarda-Nardini, L.; Piccotti, F.; Mogno, G.; Favero, L.; Manfredini, D. Age-related differences in temporomandibular disorder diagnoses. *Cranio* **2012**, *30*, 103–109. [CrossRef] [PubMed]
13. Deregibus, A.; Ferrillo, M.; Grazia Piancino, M.; Chiara Domini, M.; de Sire, A.; Castroflorio, T. Are occlusal splints effective in reducing myofascial pain in patients with muscle-related temporomandibular disorders? A randomized-controlled trial. *Turk. J. Phys. Med. Rehabil.* **2021**, *67*, 32–40. [CrossRef]
14. Moore, A.P.; Wood, G.D. The medical management of masseteric hypertrophy with botulinum toxin type A. *Br. J. Oral Maxillofac. Surg.* **1994**, *32*, 26–28. [CrossRef]
15. Lippi, L.; Ferrillo, M.; Losco, L.; Folli, A.; Marcasciano, M.; Curci, C.; Moalli, S.; Ammendolia, A.; de Sire, A.; Invernizzi, M. Aesthetic Rehabilitation Medicine: Enhancing Wellbeing beyond Functional Recovery. *Medicina* **2024**, *60*, 603. [CrossRef] [PubMed]
16. Bayrak, N.B.; Zeybek, M.; Sanlioglu, I.; Dolanmaz, D. Assessment of changes in masseter muscle by three-dimensional close-range photogrammetry after Botulinum toxin type-A injection: A case report with review of literature. *J. Pak. Med. Assoc.* **2019**, *69*, 418–422. [PubMed]
17. Khawaja, S.T. A Systematic Review: The Use of Botulinum Toxin a for the Treatment of Masseter Hypertrophy and Masticatory Myofascial Pain Associated with Bruxism. Master's Thesis, Boston University, Boston, MA, USA, 2024.
18. Chiodo, M.V.; Lisiecki, J.L.; Rohrich, R.J. Neuromodulator Finesse for Masseter Hypertrophy and Bruxism. *Plast. Reconstr. Surg.* **2024**, *153*, 726e–729e.
19. Almukhtar, R.M.; Fabi, S.G. The Masseter Muscle and Its Role in Facial Contouring, Aging, and Quality of Life: A Literature Review. *Plast. Reconstr. Surg.* **2019**, *143*, 39e–48e. [CrossRef]
20. Li, Z.; Chi, Y.; Chen, C.; Jin, L.; Huang, J.; Long, X.; Yu, N. A Comprehensive Ultrasound Evaluation Approach of Lower Facial Structure Before Masseter Muscle Botulinum Toxin Injection. *Aesthet. Surg. J.* **2023**, *43*, NP283–NP292. [CrossRef] [PubMed]
21. Graziano, P.; Orabona, G.D.; Astarita, F.; Ponzo, L.M.; Nunziata, R.; Salzano, G.; Maglitto, F.; Solari, D.; Santella, A.; Cappabianca, M.; et al. Bilateral hypertrophy of masseteric and temporalis muscles, our fifteen patients and review of literature. *Eur. Rev. Med. Pharmacol. Sci.* **2016**, *20*, 7–11.
22. Guruprasad, R.; Rishi, S.; Nair, P.P.; Thomas, S. Masseter and medial pterygoid muscle hypertrophy. *BMJ Case Rep.* **2011**, *2011*, bcr0720114557.
23. Fyfe, E.C.; Kabala, J.; Guest, P.G. Magnetic resonance imaging in the diagnosis of asymmetrical bilateral masseteric hypertrophy. *Dentomaxillofac. Radiol.* **1999**, *28*, 52–54. [CrossRef] [PubMed]
24. Boltshauser, E. Hypertrophy of temporalis muscles due to chewing gum "abuse". *J. Child. Neurol.* **1996**, *11*, 210. [CrossRef]
25. Olchowy, C.; Grzech-Leśniak, K.; Hadzik, J.; Olchowy, A.; Łasecki, M. Monitoring of Changes in Masticatory Muscle Stiffness after Gum Chewing Using Shear Wave Elastography. *J. Clin. Med.* **2021**, *10*, 2480. [CrossRef] [PubMed]
26. Park, Y.J.; Jo, Y.W.; Bang, S.I.; Kim, H.J.; Lim, S.Y.; Mun, G.H.; Hyon, W.S.; Oh, K.S. Radiofrequency volumetric reduction for masseteric hypertrophy. *Aesthet. Plast. Surg.* **2007**, *31*, 42–52. [CrossRef] [PubMed]
27. Kebede, B.; Megersa, S. Idiopathic masseter muscle hypertrophy. *Ethiop. J. Health Sci.* **2011**, *21*, 209–212.
28. Nakata, M. Masticatory function and its effects on general health. *Int. Dent. J.* **1998**, *48*, 540–548, Correction in *Int. Dent. J.* **1999**, *49*, 3. [CrossRef] [PubMed]
29. Kumar, A.; Almotairy, N.; Merzo, J.J.; Wendin, K.; Rothenberg, E.; Grigoriadis, A.; Sandborgh-Englund, G.; Trulsson, M. Chewing and its influence on swallowing, gastrointestinal and nutrition-related factors: A systematic review. *Crit. Rev. Food Sci. Nutr.* **2023**, *63*, 11987–12017. [CrossRef]
30. Sannomya, E.K.; Gonçalves, M.; Cavalcanti, M.P. Masseter muscle hypertrophy: Case report. *Braz. Dent. J.* **2006**, *17*, 347–350. [CrossRef] [PubMed]
31. Nikolis, A.; Enright, K.M.; Rudolph, C.; Cotofana, S. Temporal volume increase after reduction of masseteric hypertrophy utilizing incobotulinumtoxin type A. *J. Cosmet. Dermatol.* **2020**, *19*, 1294–1300. [CrossRef] [PubMed]
32. Satoh, K.; Yamaguchi, T.; Komatsu, K.; Inoue, N.; Minowa, K.; Kanayama, T.; Yoshida, S.; Ohata, N. Analyses of muscular activity, energy metabolism, and muscle fiber type composition in a patient with bilateral masseteric hypertrophy. *Cranio* **2001**, *19*, 294–301. [CrossRef]
33. Okeson, J.P. *Management of Temporomandibular Disorders and Occlusion*, 8th ed.; Elsevier Health Sciences: St. Louis, MO, USA, 2020.
34. Stål, P.; Erikssonm, P.O.; Schiaffino, S.; Butler-Browne, G.S.; Thornell, L.E. Differences in myosin composition between human oro-facial, masticatory and limb muscles: Enzyme-, immunohisto- and biochemical studies. *J. Muscle Res. Cell Motil.* **1994**, *15*, 517–534. [CrossRef] [PubMed]
35. Korfage, J.A.; Koolstra, J.H.; Langenbach, G.E.; van Eijden, T.M. Fiber-type composition of the human jaw muscles—(Part 1) origin and functional significance of fiber-type diversity. *J. Dent. Res.* **2005**, *84*, 774–783. [CrossRef] [PubMed]

36. Kato, Y.; Hoshino, T.; Ogawa, Y.; Sugahara, K.; Katakura, A. Aging-Related Metabolome Analysis of the Masseter Muscle in Senescence-Accelerated Mouse-Prone 8. *Int. J. Mol. Sci.* **2024**, *25*, 9684. [CrossRef] [PubMed]
37. Eriksson, P.O.; Thornell, L.E. Histochemical and morphological muscle-fibre characteristics of the human masseter, the medial pterygoid and the temporal muscles. *Arch. Oral Biol.* **1983**, *28*, 781–795. [CrossRef] [PubMed]
38. Niszezak, C.M.; Sonza, A.; Garrett, A.; Santos, G.M. Muscle oxygenation and pain in different types of temporomandibular disorders. *Clin. Oral Investig.* **2024**, *28*, 410. [CrossRef]
39. Zhu, Y.; Zhu, J.; Yin, D.; Liu, Y. Improved stomatognathic model for highly realistic finite element analysis of temporomandibular joint biomechanics. *J. Mech. Behav. Biomed. Mater.* **2024**, *160*, 106780. [CrossRef] [PubMed]
40. Baricich, A.; Picelli, A.; Santamato, A.; Carda, S.; de Sire, A.; Smania, N.; Cisari, C.; Invernizzi, M. Safety Profile of High-Dose Botulinum Toxin Type A in Post-Stroke Spasticity Treatment. *Clin. Drug Investig.* **2018**, *38*, 991–1000. [CrossRef] [PubMed]
41. Neumeyer, T.; Schiffler, B.; Maier, E.; Lang, A.E.; Aktories, K.; Benz, R. Clostridium botulinum C2 toxin. Identification of the binding site for chloroquine and related compounds and influence of the binding site on properties of the C2II channel. *J. Biol. Chem.* **2008**, *283*, 3904–3914. [CrossRef] [PubMed]
42. Raciti, L.; Raciti, G.; Ammendolia, A.; de Sire, A.; Onesta, M.P.; Calabrò, R.S. Improving Spasticity by Using Botulin Toxin: An Overview Focusing on Combined Approaches. *Brain Sci.* **2024**, *14*, 631. [CrossRef] [PubMed]
43. Tam, E.; Choo, J.P.S.; Rao, P.; Webb, W.R.; Carruthers, J.D.A.; Rahman, E. A Systematic Review on the Effectiveness and Safety of Combining Biostimulators with Botulinum Toxin, Dermal Fillers, and Energy-Based Devices. *Aesthet. Plast. Surg.* **2024**. [CrossRef]
44. Dressler, D.; Benecke, R. Pharmacology of therapeutic botulinum toxin preparations. *Disabil. Rehabil.* **2007**, *29*, 1761–1768. [CrossRef]
45. Hambleton, P. Clostridium botulinum toxins: A general review of involvement in disease, structure, mode of action and preparation for clinical use. *J. Neurol.* **1992**, *239*, 16–20. [CrossRef] [PubMed]
46. Baricich, A.; Picelli, A.; Carda, S.; Smania, N.; Cisari, C.; Santamato, A.; de Sire, A.; Invernizzi, M. Electrical stimulation of antagonist muscles after botulinum toxin type A for post-stroke spastic equinus foot. A randomized single-blind pilot study. *Ann. Phys. Rehabil. Med.* **2019**, *62*, 214–219. [CrossRef] [PubMed]
47. Lippi, L.; de Sire, A.; Folli, A.; D'abrosca, F.; Grana, E.; Baricich, A.; Carda, S.; Invernizzi, M. Multidimensional Effectiveness of Botulinum Toxin in Neuropathic Pain: A Systematic Review of Randomized Clinical Trials. *Toxins* **2022**, *14*, 308. [CrossRef] [PubMed]
48. Burstein, R.; Blumenfeld, A.M.; Silberstein, S.D.; Manack Adams, A.; Brin, M.F. Mechanism of Action of OnabotulinumtoxinA in Chronic Migraine: A Narrative Review. *Headache* **2020**, *60*, 1259–1272. [CrossRef] [PubMed]
49. Ho, T.W.; Edvinsson, L.; Goadsby, P.J. CGRP and its receptors provide new insights into migraine pathophysiology. *Nat. Rev. Neurol.* **2010**, *6*, 573–582. [CrossRef]
50. Ramachandran, R.; Lam, C.; Yaksh, T.L. Botulinum toxin in migraine: Role of transport in trigemino-somatic and trigemino-vascular afferents. *Neurobiol. Dis.* **2015**, *79*, 111–122. [CrossRef]
51. Cernuda-Morollón, E.; Ramón, C.; Martínez-Camblor, P.; Serrano-Pertierra, E.; Larrosa, D.; Pascual, J. OnabotulinumtoxinA decreases interictal CGRP plasma levels in patients with chronic migraine. *Pain* **2015**, *156*, 820–824. [CrossRef]
52. Ceruti, S.; Villa, G.; Fumagalli, M.; Colombo, L.; Magni, G.; Zanardelli, M.; Fabbretti, E.; Verderio, C.; van den Maagdenberg, A.M.; Nistri, A.; et al. Calcitonin gene-related peptide-mediated enhancement of purinergic neuron/glia communication by the algogenic factor bradykinin in mouse trigeminal ganglia from wild-type and R192Q Cav2.1 Knock-in mice: Implications for basic mechanisms of migraine pain. *J. Neurosci.* **2011**, *31*, 3638–3649. [CrossRef] [PubMed]
53. Lu, Y.; Jiang, Q.; Yu, L.; Lu, Z.-Y.; Meng, S.-P.; Su, D.; Burnstock, G.; Ma, B. 17β-estradiol rapidly attenuates P2X3 receptor-mediated peripheral pain signal transduction via ERα and GPR30. *Endocrinology* **2013**, *154*, 2421–2433. [CrossRef] [PubMed]
54. Zhang, X.; Strassman, A.M.; Novack, V.; Brin, M.F.; Burstein, R. Extracranial injections of botulinum neurotoxin type A inhibit intracranial meningeal nociceptors' responses to stimulation of TRPV1 and TRPA1 channels: Are we getting closer to solving this puzzle? *Cephalalgia* **2016**, *36*, 875–886. [CrossRef]
55. Yoshida, K. Botulinum Toxin Therapy for Oromandibular Dystonia and Other Movement Disorders in the Stomatognathic System. *Toxins* **2022**, *14*, 282. [CrossRef]
56. Pihut, M. *Efficacy of Prosthetic and Pharmacological Relaxation of Masseter Muscles as Alternative Methods of Treatment of Masticatory Organ Dysfunction*; Monograph: Kraków, Poland, 2012.
57. Ghavimi, M.A.; Yazdani, J.; Afzalimehr, A.; Ghoreyshizadeh, A.; Dehnad, S.V. Effect of injection of botulinum toxin on decreasing the symptoms and signs of masticatory muscles in patients with temporomandibular dysfunction. *J. Dent. Res. Dent. Clin. Dent. Prospects* **2019**, *13*, 128–132. [CrossRef]
58. Rossetto, O.; Pirazzini, M.; Montecucco, C. Botulinum neurotoxins: Genetic, structural and mechanistic insights. *Nat. Rev. Microbiol.* **2014**, *12*, 535–549. [CrossRef] [PubMed]
59. Kassir, M.; Babaei, M.; Hasanzadeh, S.; Rezaei Tavirani, M.; Razzaghi, Z.; Robati, R.M. Botulinium toxin applications in the lower face and neck: A comprehensive review. *J. Cosmet. Dermatol.* **2024**, *23*, 1205–1216. [CrossRef] [PubMed]

60. Kundu, N.; Kothari, R.; Shah, N.; Sandhu, S.; Tripathy, D.M.; Galadari, H.; Gold, M.H.; Goldman, M.P.; Kassir, M.; Schepler, H.; et al. Efficacy of botulinum toxin in masseter muscle hypertrophy for lower face contouring. *J. Cosmet. Dermatol.* **2022**, *21*, 1849–1856. [CrossRef]
61. Ozdemir Cetinkaya, P.; Karaosmanoglu, N.; Özkesici Kurt, B.; Aksu Cerman, A.; Altunay, I.K. Functional and esthetic effects of botulinum toxin injection into the masseter muscles: Evaluation of 80 patients from a dermatological perspective. *Int. J. Dermatol.* **2025**, *64*, 149–154. [CrossRef] [PubMed]
62. Chang, C.-S.; Lin, S.; Wallace, C.G.; Hsiao, Y.-C.; Lin, C.-M.; Kang, G.C.-W.; Chen, Z.-C.; Chen, P.K.-T.; Lo, L.-J.; Chen, Y.-R.; et al. Masseter muscle volume changes evaluated by 3-dimensional computed tomography after repeated botulinum toxin a injections in patients with square facial morphology. *Ann. Plast. Surg.* **2019**, *82* (Suppl. S1), S29–S32. [CrossRef]
63. Lee, H.H.; Kim, S.T.; Lee, K.J.; Baik, H.S. Effect of a second injection of botulinum toxin on lower facial contouring, as evaluated using 3-dimensional laser scanning. *Dermatol. Surg.* **2015**, *41*, 439–444. [CrossRef]
64. Ryoo, H.J.; Kwon, H.; Choi, J.S.; Sohn, B.S.; Yoo, J.Y.; Shim, H.S. Prospective Analysis of the Effectiveness of Targeted Botulinum Toxin Type A Injection Using an Ultrasound-Guided Single-Point Injection Technique for Lower Face Contouring. *J. Clin. Med.* **2024**, *13*, 5337. [CrossRef]
65. Wan, J.; Kim, J.S.; Park, Y.; Park, S.Y.; Koppert, E.; Kim, H.J.; Yi, K.H. Novel single-entry point injection technique for masseter hypertrophy treatment using botulinum neurotoxin based on patient-reported comfort. *J. Cosmet. Dermatol.* **2024**, *23*, 3539–3543. [CrossRef] [PubMed]
66. Huang, D.-W.; Lai, C.-Y.; Chen, J.-E.; Yi, C.-C.; Chen, Y.-H.; Wang, C.-H.; Chen, S.-G. Three-Dimensional Photography for Evaluating the Effectiveness of Botulinum Toxin Injection for Masseter Hypertrophy. *Aesthet. Plast. Surg.* **2024**, *48*, 4065–4076. [CrossRef] [PubMed]
67. Xie, Y.; Zhou, J.; Li, H.; Cheng, C.; Herrler, T.; Li, Q. Classification of masseter hypertrophy for tailored botulinum toxin type A treatment. *Plast. Reconstr. Surg.* **2014**, *134*, 209e–218e. [CrossRef] [PubMed]
68. Nikolis, A.; Enright, K.M.; Masouri, S.; Bernstein, S.; Antoniou, C. Prospective evaluation of incobotulinumtoxinA in the management of the masseter using two different injection techniques. *Clin. Cosmet. Investig. Dermatol.* **2018**, *11*, 347–356. [CrossRef] [PubMed]
69. Chang, C.S.; Bergeron, L.; Yu, C.C.; Chen, P.K.; Chen, Y.R. Mandible changes evaluated by computed tomography following botulinum toxin A injections in square-faced patients. *Aesthet. Plast. Surg.* **2011**, *35*, 452–455. [CrossRef]
70. Kim, N.H.; Chung, J.H.; Park, R.H.; Park, J.B. The use of botulinum toxin type A in aesthetic mandibular contouring. *Plast. Reconstr. Surg.* **2005**, *115*, 919–930. [CrossRef]
71. Popescu, M.N.; Beiu, C.; Iliescu, C.A.; Racoviță, A.; Berteanu, M.; Iliescu, M.G.; Stănescu, A.M.A.; Radaschin, D.S.; Popa, L.G. Ultrasound-Guided Botulinum Toxin-A Injections into the Masseter Muscle for Both Medical and Aesthetic Purposes. *Toxins* **2024**, *16*, 413. [CrossRef] [PubMed]
72. de Souza Nobre, B.B.; Rezende, L.; Barbosa Câmara-Souza, M.; Sanchez-Ayala, A.; Blass, R.; Carbone, A.C.; Manso, A.C.; Ernberg, M.; Christidis, N.; De la Torre Canales, G. Exploring botulinum toxin's impact on masseter hypertrophy: A randomized, triple-blinded clinical trial. *Sci. Rep.* **2024**, *14*, 14522. [CrossRef] [PubMed]
73. Chen, Y.; Tsai, C.H.; Bae, T.H.; Huang, C.Y.; Chen, C.; Kang, Y.N.; Chiu, W.K. Effectiveness of Botulinum Toxin Injection on Bruxism: A Systematic Review and Meta-analysis of Randomized Controlled Trials. *Aesthet. Plast. Surg.* **2023**, *47*, 775–790. [CrossRef]
74. Rathod, N.N.; John, R.S. Botulinum Toxin Injection for Masseteric Hypertrophy Using 6 Point Injection Technique—A Case Report. Proposal of a Clinical Technique to Quantify Prognosis. *Clin. Cosmet. Investig. Dent.* **2023**, *15*, 45–49. [CrossRef]
75. Chirico, F.; Bove, P.; Fragola, R.; Cosenza, A.; De Falco, N.; Giudice, G.L.; Audino, G.; Rauso, G.M. Biphasic injection for masseter muscle reduction with botulinum toxin. *Appl. Sci.* **2021**, *11*, 6478. [CrossRef]
76. Seok, J.; Koh, Y.G.; Hong, J.K.; Yun, S.H.; Kim, D.H.; Son, H.S.; Choi, S.Y.; Yoo, K.H.; Lee, Y.W.; Kim, B.J. Efficacy and Safety of PrabotulinumtoxinA in Subjects With Benign Masseteric Hypertrophy: A Double-Blind, Randomized, Placebo-Controlled, Multicenter, Phase 3 Trial and Open-Label Extension Study. *Dermatol. Surg.* **2024**, *50*, 527–533. [CrossRef] [PubMed]
77. De la Torre Canales, G.; Poluha, R.L.; Bonjardim, L.R.; Ernberg, M.; Conti, P.C.R. Botulinum toxin-A effects on pain, somatosensory and psychosocial features of patients with refractory masticatory myofascial pain: A randomized double-blind clinical trial. *Sci. Rep.* **2024**, *14*, 4201.
78. Gil-Martinez, A.; Paris-Alemany, A.; López-De-Uralde-Villanueva, I.; La Touche, R. Management of Pain in Patients with Temporomandibular Disorder (TMD): Challenges and Solutions. *J. Pain Res.* **2018**, *11*, 571–587. [CrossRef]
79. Kaya, D.I.; Ataoglu, H. Botulinum Toxin Treatment of Temporomandibular Joint Pain in Patients with Bruxism: A Prospective and Randomized Clinical Study. Niger. *J. Clin. Pract.* **2021**, *24*, 412–417. [CrossRef]
80. Hosgor, H.; Altindis, S. Efficacy of botulinum toxin in the management of temporomandibular myofascial pain and sleep bruxism. *J. Korean Assoc. Oral Maxillofac. Surg.* **2020**, *46*, 335–340. [CrossRef]

81. Koo, H.J.; Hu, H.; Kim, W.; Kim, J.S.; Kim, H.J.; Yi, K.H. Do repetitive botulinum neurotoxin injections induce muscle fibrosis? Sonographic observation of the masseter muscle. *J. Cosmet. Dermatol.* **2024**, *23*, 434–440. [CrossRef] [PubMed]
82. Mandel, L.; Tharakan, M. Treatment of unilateral masseteric hypertrophy with botulinum toxin: Case report. *J. Oral Maxillofac. Surg.* **1999**, *57*, 1017–1019. [CrossRef] [PubMed]
83. Tartaro, G.; Rauso, R.; Santagata, M.; Santillo, V.; Itro, A. Lower facial contouring with botulinum toxin type A. *J. Craniofacial Surg.* **2008**, *19*, 1613–1617. [CrossRef] [PubMed]
84. Baldwin, M.C.; Liu, Z.J.; Rafferty, K.L.; Keith, A.; Tamasas, B.; Kaiyala, K.; Herring, S.W. Botulinum toxin in the masseter muscle: Lingering effects of denervation. *Anat. Rec.* **2022**, *305*, 1215–1230. [CrossRef] [PubMed]
85. Mkhitaryan, L.; Alcolea, J.M. Prospective Clinical Study and Ultrasound Assessment in Patients with Bruxism Treated with Botulinum Toxin. *Aesthetic Med.* **2020**, *6*, 25–34.
86. Lee, H.J.; Kang, I.W.; Seo, K.K.; Choi, Y.J.; Kim, S.T.; Hu, K.S.; Kim, H.J. The Anatomical Basis of Paradoxical Masseteric Bulging after Botulinum Neurotoxin Type a Injection. *Toxins* **2017**, *9*, 14. [CrossRef] [PubMed]
87. Ho, W.W.S.; Chan, L.; Corduff, N.; Lau, W.-T.; Martin, M.U.; Tay, C.M.; Wang, S.; Wu, R. Addressing the Real-World Challenges of Immunoresistance to Botulinum Neurotoxin A in Aesthetic Practice: Insights and Recommendations from a Panel Discussion in Hong Kong. *Toxins* **2023**, *15*, 456. [CrossRef]
88. Shi, J.; Li, C.; Zhou, J.; Guo, X.; Li, G.; You, M. An Ultrasonographic Analysis of the Deep Inferior Tendon in the Masseter Muscle: Implications for Botulinum Toxin Injections. *Toxins* **2024**, *16*, 391. [CrossRef]
89. Shim, Y.J.; Lee, H.J.; Park, K.J.; Kim, H.T.; Hong, I.H.; Kim, S.T. Botulinum Toxin Therapy for Managing Sleep Bruxism: A Randomized and Placebo-Controlled Trial. *Toxins* **2020**, *12*, 168. [CrossRef] [PubMed]
90. Zhang, L.D.; Liu, Q.; Zou, D.R.; Yu, L.F. Occlusal force characteristics of masseteric muscles after intramuscular injection of botulinum toxin A(BTX—A)for treatment of temporomandibular disorder. *Br. J. Oral Maxillofac. Surg.* **2016**, *54*, 736–740. [CrossRef]
91. Diracoglu, D.; Sahbaz, T.; Alptekin, K.; Dogan, N. Effects of ultrasound-assisted botulinum neurotoxin-A injection in patients with bruxism and masseter hypertrophy. *Turk. J. Phys. Med. Rehabil.* **2021**, *67*, 351–356. [CrossRef]
92. Park, Y.; Ku, S.K.; Lee, D.H.; Kim, S.T. Combined Effects of Botulinum Toxin Injection and Oral Appliance Therapy on Lower Facial Contouring: A Randomized Controlled Trial. *J. Clin. Med.* **2022**, *11*, 4092. [CrossRef] [PubMed]
93. Lee, S.J.; McCall, W.D., Jr.; Kim, Y.K.; Chung, S.C.; Chung, J.W. Effect of botulinum toxin injection on nocturnal bruxism: A randomized controlled trial. *Am. J. Phys. Med. Rehabil.* **2010**, *89*, 16–23. [CrossRef] [PubMed]
94. Jung, B.K.; Park, H.; Cheon, Y.W.; Yun, I.S.; Choi, J.-W.; Kim, H.J.; Lee, M.Y.; Kang, B.S.; Kang, T.J. Clinical investigation of botulinum toxin (prabotulinumtoxin A) for bruxism related to masseter muscle hypertrophy: A prospective study. *J. Cranio-Maxillofac. Surg.* **2023**, *51*, 332–337. [CrossRef] [PubMed]
95. Alwayli, H.; Abdulrahman, B.I.; Rastogi, S. Does botulinum toxin have any role in the management of chronic pain associated with bruxism? *Cranio* **2024**, *42*, 215–222. [CrossRef] [PubMed]
96. Guo, Y.; Diao, X.; Dong, D.; Xia, W.; Liu, T.; Zhou, Y.; Zhu, J.; Chen, L.; Chen, Y. Effects of Two Botulinum Toxin Type a Evaluated by Shear Wave Elastography and Electromyographic Measurements of Masseter Reduction. *J. Craniofacial Surg.* **2022**, *33*, 1450–1453. [CrossRef] [PubMed]
97. Sendra, L.A.; Azeredo Alves Antunes, L.; Barboza, E.P. Use of botulinum neurotoxin Type A in the management of primary bruxism in adults: An updated systematic review. *J. Prosthet. Dent.* **2024**, *132*, 93–99. [CrossRef] [PubMed]
98. Hong, J.Y.; Jeong, G.J.; Kwon, T.R.; Kim, J.H.; Li, K.; Kim, B.J. Efficacy and Safety of a Novel Botulinum Toxin A for Masseter Reduction: A Randomized, Double-Blind, Placebo-Controlled, Optimal Dose-Finding Study. *Dermatol. Surg.* **2021**, *47*, e5–e9. [CrossRef] [PubMed]
99. Asutay, F.; Atalay, Y.; Asutay, H.; Acar, A.H. The Evaluation of the Clinical Effects of Botulinum Toxin on Nocturnal Bruxism. *Pain Res. Manag.* **2017**, *2017*, 6264146. [CrossRef] [PubMed]
100. Sidebottom, A.J.; Patel, A.A.; Amin, J. Botulinum injection for the management of myofascial pain in the masticatory muscles. A prospective outcome study. *Br. J. Oral Maxillofac. Surg.* **2013**, *51*, 199–205. [CrossRef] [PubMed]
101. Guarda-Nardini, L.; Manfredini, D.; Salamone, M.; Salmaso, L.; Tonello, S.; Ferronato, G. Efficacy of botulinum toxin in treating myofascial pain in bruxers: A controlled placebo pilot study. *Cranio* **2008**, *26*, 126–135. [CrossRef]
102. Li, K.; Tan, K.; Yacovelli, A.; Bi, W.G. Effect of botulinum toxin type A on muscular temporomandibular disorder: A systematic review and meta-analysis of randomized controlled trials. *J. Oral Rehabil.* **2024**, *51*, 886–897. [CrossRef]
103. Delcanho, R.; Val, M.; Guarda Nardini, L.; Manfredini, D. Botulinum Toxin for Treating Temporomandibular Disorders: What is the Evidence? *J. Oral Facial Pain Headache* **2022**, *36*, 6–20. [CrossRef]
104. Machado, D.; Martimbianco, A.L.C.; Bussadori, S.K.; Pacheco, R.L.; Riera, R.; Santos, E.M. Botulinum toxin type a for painful temporomandibular disorders: Systematic review and meta-analysis. *J. Pain* **2020**, *21*, 281–293. [CrossRef] [PubMed]

105. Ramos-Herrada, R.M.; Arriola-Guillén, L.E.; Atoche-Socola, K.J.; Bellini-Pereira, S.A.; Castillo, A.A.D. Effects of botulinum toxin in patients with myofascial pain related to temporomandibular joint disorders: A systematic review. *Dent. Med. Probl.* **2022**, *59*, 271–280. [CrossRef]
106. Thambar, S.; Kulkarni, S.; Armstrong, S.; Nikolarakos, D. Botulinum toxin in the management of temporomandibular disorders: A systematic review. *Br. J. Oral Maxillofac. Surg.* **2020**, *58*, 508–519. [CrossRef]
107. Signorini, M.; Fundarò, S.P.; Bertossi, D.; Cavallini, M.; Cirillo, P.; Natuzzi, G.; Quartucci, S.; Sciuto, C.; Patalano, M.; Trocchi, G. OnabotulinumtoxinA from lines to facial reshaping: A new Italian consensus report. *J. Cosmet. Dermatol.* **2022**, *21*, 550–563. [CrossRef]
108. Zhang, S.; Zhao, H.; Liu, C.; Gao, X.; Hao, L. 3D Assessment of Mandibular Margin Morphological Change after BTX-A Injection for Masseter Hypertrophy: A Retrospective Study. *Aesthet. Plast. Surg.* **2024**. [CrossRef]
109. Rauso, R.; Lo Giudice, G.; Tartaro, G.; Zerbinati, N.; Nicoletti, G.F.; Fragola, R. Botulinum toxin type A injections for masticatory muscles hypertrophy: A systematic review. *J. Cranio-Maxillofac. Surg.* **2022**, *50*, 7–18. [CrossRef]
110. Kim, D.-H.; Hong, H.-S.; Won, S.-Y.; Kim, H.-J.; Hu, K.-S.; Choi, J.-H.; Kim, H.-J. Intramuscular nerve distribution of the masseter muscle as a basis for botulinum toxin injection. *J. Craniofacial Surg.* **2010**, *21*, 588–591. [CrossRef] [PubMed]
111. Peng, H.P.; Peng, J.H. Complications of botulinum toxin injection for masseter hypertrophy: Incidence rate from 2036 treatments and summary of causes and preventions. *J. Cosmet. Dermatol.* **2018**, *17*, 33–38. [CrossRef] [PubMed]
112. Ferrari, A.; Manca, M.; Tugnoli, V.; Alberto, L. Pharmacological differences and clinical implications of various botulinum toxin preparations: A critical appraisal. *Funct. Neurol.* **2018**, *33*, 7–18. [CrossRef]
113. Yeh, Y.T.; Peng, J.H.; Peng, H.-P. Literature review of the adverse events associated with botulinum toxin injection for the masseter muscle hypertrophy. *J. Cosmet. Dermatol.* **2018**, *17*, 675–687. [CrossRef] [PubMed]
114. Shome, D.; Khare, S.; Kapoor, R. Efficacy of Botulinum Toxin in Treating Asian Indian Patients with Masseter Hypertrophy: A 4-Year Follow-Up Study. *Plast. Reconstr. Surg.* **2019**, *144*, 390e–396e. [CrossRef]
115. Guida, S. Neurotoxin in the Lower Third of the Face. *Dermatol. Clin.* **2023**, *42*, 63–67.
116. Bae, J.H.; Choi, D.Y.; Lee, J.G. The risorius muscle: Anatomic considerations with reference to botulinum neurotoxin injection for masseteric hypertrophy. *Dermatol. Surg.* **2014**, *40*, 1334–1339. [CrossRef] [PubMed]
117. Frevert, J.; Dressler, D. Complexing proteins in botulinum toxin type A drugs: A help or a hindrance? *Biologics* **2010**, *4*, 325–332. [CrossRef]
118. Dressler, D. Botulinum toxin drugs: Brief history and outlook. *J. Neural Transm.* **2015**, *119*, 877–879. [CrossRef] [PubMed]
119. Yiannakopoulou, E. Serious and long-term adverse events associated with the therapeutic and cosmetic use of botulinum toxin. *Pharmacology* **2015**, *95*, 65–69. [CrossRef] [PubMed]
120. Rahman, E.; Carruthers, J.D.A. Immunogenicity of Botulinum Toxin A: Insights. *Dermatol. Surg.* **2024**, *50*, S117–S126. [CrossRef] [PubMed]
121. Borodic, G. Botulinum toxin, immunologic considerations with long-term repeated use, with emphasis on cosmetic applications. *Facial Plast. Surg. Clin. N. Am.* **2007**, *15*, 11–16. [CrossRef] [PubMed]
122. Erdil, D.; Bagis, N.; Eren, H.; Camgoz, M.; Orhan, K. The Evaluation of the Relationship between Changes in Masseter Muscle Thickness and Tooth Clenching Habits of Bruxism Patients Treated with Botulinum Toxin A. *J. Med. Ultrasound* **2022**, *31*, 22–28. [CrossRef]
123. Moussa, M.S.; Bachour, D.; Komarova, S.V. Adverse effect of botulinum toxin-A injections on mandibular bone: A systematic review and meta-analysis. *J. Oral Rehabil.* **2024**, *51*, 404–415. [CrossRef]
124. Hong, S.W.; Kang, J.H. Decreased mandibular cortical bone quality after botulinum toxin injections in masticatory muscles in female adults. *Sci. Rep.* **2020**, *10*, 3623. [CrossRef] [PubMed]
125. Tsai, C.Y.; Shyr, Y.M.; Chiu, W.C.; Lee, C.M. Bone changes in the mandible following botulinum neurotoxin injections. *Eur. J. Orthod.* **2011**, *33*, 32–138. [CrossRef]
126. Kahn, A.; Kün-Darbois, J.D.; Bertin, H.; Corre, P.; Chappard, D. Mandibular bone effects of botulinum toxin injections in masticatory muscles in adult. *Oral Surg. Oral Med. Oral Pathol. Oral Radiol.* **2020**, *129*, 100–108. [CrossRef]
127. Li, Y.; Zheng, Q.; Lin, J.; Su, X.; Zhuang, J.; Wei, Q.; Hu, J. Mild Allergic Reactions after Botulinum Toxin Injection: A Case Series and Literature Review. *Plast. Reconstr. Surg. Glob. Open* **2024**, *12*, e5845. [CrossRef]

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